$$= \frac{\dfrac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\cos\alpha\cos\beta}}{\dfrac{\cos\alpha\cos\beta - \sin\alpha\sin\beta}{\cos\alpha\cos\beta}}$$

$$\text{Hence, } \tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$

Identity 3.6: $\tan(\alpha-\beta) = \dfrac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$

Proof. This result may be proved by writing $\tan(\alpha-\beta) = \tan[\alpha + (-\beta)]$ and using the Identity 3.5.

(i) Ptolemy (CE 100-170) treated the chord of an angle as his basic trigonometric function and proved the theorem: In a cyclic quadrilateral, the product of diagonals equals the sum of the products of opposite sides. That is, in a cyclic quadrilateral $ABCD$,

$$(AC)(BD) = (AB)(CD) + (AD)(BC).$$

Using this theorem, one can prove Sum and Difference identities. Hence, these identites are known as Ptolemy's sum and difference formulas.

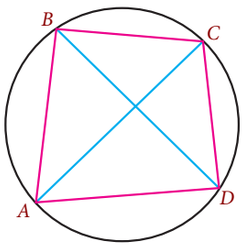


Figure 3.16

(ii) Observe that $\cos(\alpha \pm \beta) \neq \cos\alpha \pm \cos\beta$. Similarly we can observe for other trigonometric functions also.

(iii) When $\alpha = \beta$, $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$ implies $\sin 0 = 0$, which we have already established.

(iv) When $\alpha = \dfrac{\pi}{2}$ and $\beta = \theta$, $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$ gives $\sin\left(\dfrac{\pi}{2} - \theta\right) = \cos\theta$, which we have already proved.

(v) We can find the trigonometric function values of a given angle, if we can break it up into sum or difference of two of the special angles. For example, we can evaluate $\tan 75^\circ$ as $\tan(45^\circ + 30^\circ)$ and $\cos 135^\circ$ as $\cos(180^\circ - 45^\circ)$.

Example 3.15 Find the values of (i) $\cos 15^\circ$ and (ii) $\tan 165^\circ$.

Solution:

(i) Now, $\cos 15^\circ = \cos(45^\circ - 30^\circ) = \cos 45^\circ\cos 30^\circ + \sin 45^\circ\sin 30^\circ$

$$= \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}\frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

Also, note that $\sin 75^\circ = \dfrac{\sqrt{3}+1}{2\sqrt{2}}$ [try yourself]

(ii) Now, $\tan 165^\circ = \tan(120^\circ + 45^\circ) = \dfrac{\tan 120^\circ + \tan 45^\circ}{1 - \tan 120^\circ\tan 45^\circ}$

But, $\tan 120^\circ = \tan(90^\circ + 30^\circ) = -\cot 30^\circ = -\sqrt{3}$ and $\tan 45^\circ = 1$

Thus, $\tan 165^\circ = \dfrac{1-\sqrt{3}}{1+\sqrt{3}}$.